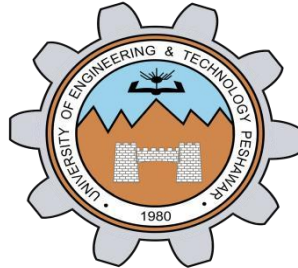


LAB 1

Introduction to Embedded Systems & Arduino Environment



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CSE-420L

Embedded Systems Lab

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“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work”

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Introduction to Embedded Systems & Arduino Environment

Objectives:

- To understand the basics of Embedded Systems.
- To learn the Arduino IDE environment.
- To write, compile and upload a simple Arduino program. □ To blink an LED using Arduino.

Apparatus / Components:

- TINKERCAD
- Arduino Uno

Theory:

An embedded system is a microcontroller-based system designed to perform a specific task.

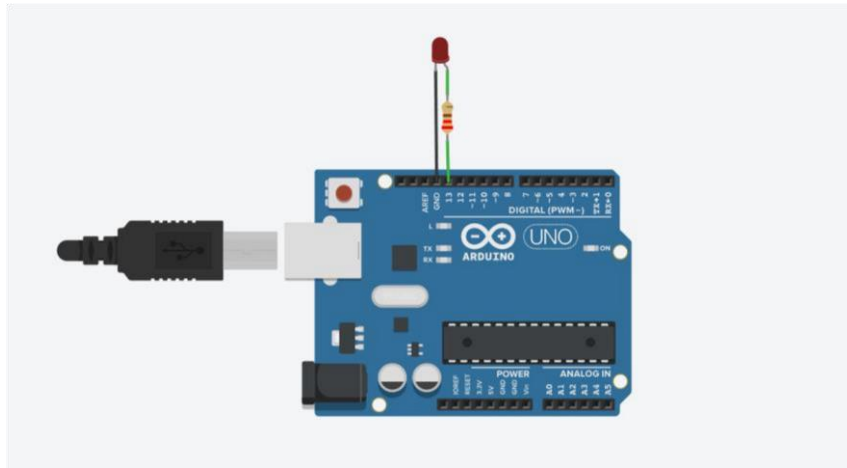
The Arduino Uno is based on the ATmega328P microcontroller. It has digital input/output pins that can be programmed using the Arduino IDE.

In this experiment:

- The LED is connected to a digital output pin.
- `pinMode()` configures the pin as OUTPUT.
- `digitalWrite()` sets the pin HIGH (5V) or LOW (0V).
- `delay()` pauses the program for a specified time in milliseconds.

The program continuously turns the LED ON and OFF every 1 second, creating a blinking effect.

CIRCUIT DIAGRAM:



Algorithm:

1. Start
2. Set LED pin as OUTPUT
3. Turn LED ON
4. Wait for 1 second
5. Turn LED OFF
6. Wait for 1 second
7. Repeat steps 3–6 continuously

Code:

```
void setup()
{
  pinMode(LED_BUILTIN, OUTPUT);
}
```

```
void loop()
{
  digitalWrite(LED_BUILTIN, HIGH);
  delay(1000);

  digitalWrite(LED_BUILTIN, LOW);
  delay(1000);
}
```

Observations:

- The LED turns ON for 1 second.
- The LED turns OFF for 1 second.
- The process repeats continuously.
- Blinking time interval = 2 seconds (1 sec ON + 1 sec OFF).

Conclusion:

The basic structure of an embedded system was studied using Arduino Uno.

The LED blinking experiment demonstrated digital output control and timing functions in Arduino.